

Bambu Lab LED Status Bar

(DIY version using Home Assistant + ESP32 + WLED or MQTT)

Please check the Wiring guide available on:

<https://makerworld.com/en/models/2172105-bambu-lab-p2s-led-progress-status-bar>

Standalone MQTT Version

The standalone version communicates directly with your Bambu Lab printer over MQTT, without requiring Home Assistant or WLED.

The ESP32 runs its own firmware, connects to your printer over your local network and controls the LED strip automatically. It also includes a web interface for configuring Wi-Fi, connecting to your printer, customising LED colours and adjusting brightness.

No coding is required. You can flash the firmware directly from your browser and complete the setup using your phone or computer.

Requirements

Before starting, make sure you have:

- An ESP32-WROOM-32 development board.
- A compatible addressable LED strip, such as the 5V WS2811 COB RGB strip used in this project.
- A Bambu Lab printer with LAN Mode enabled and accessible over your local network.
- A Wi-Fi network accessible to both the ESP32 and printer.
- A USB data cable and Chrome or Edge for flashing the ESP32.

You'll also need your printer's IP address, serial number and LAN Access Code. The setup wizard can automatically discover your printer, so you might not need to enter the IP address and serial number manually.

No Home Assistant, WLED or additional server is required.

Printer Compatibility

P2S	☒ Tested
X1C	☐ Expected to work
P1S	☐ Expected to work
Other Bambu models	☐ Not confirmed yet

The standalone firmware was developed around my Bambu Lab P2S. Other models that provide compatible local MQTT status information may also work, but I haven't fully tested them yet.

If you test the firmware with another Bambu Lab printer, please share your results in the MakerWorld comments so I can keep the compatibility information updated.

How It Works

The ESP32 connects directly to the printer's local MQTT service and receives information about its current state and print progress.

The firmware automatically selects the appropriate LED animation for the following states:

Printer state	LED effect
Idle	Plasmoid
Calibrating	Loading
Printing	Percent
Paused	Slow Fade
Finished	Green Fade
Error	Fast Fade

Heating, homing, bed levelling and filament changes are currently grouped into the Calibrating state.

During printing, the strip displays a visual progress bar. With the default configuration of 10 LED sections, each section represents approximately 10% of the print.

The firmware also reduces brightness when the printer's chamber light is off. After an extended period of inactivity, it dims the LEDs further.

Unlike the Home Assistant version, you don't need to create WLED presets or import any automations.

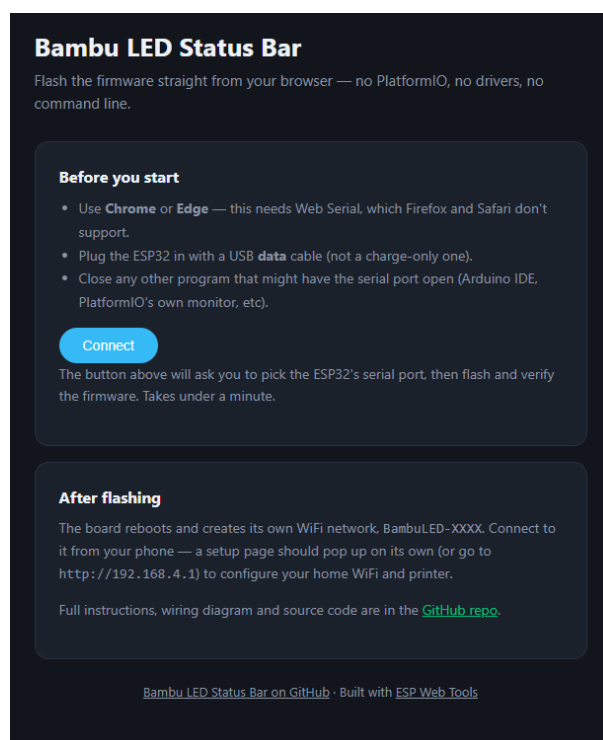
1. Flashing the Firmware

The easiest way to install the firmware is through the web installer.

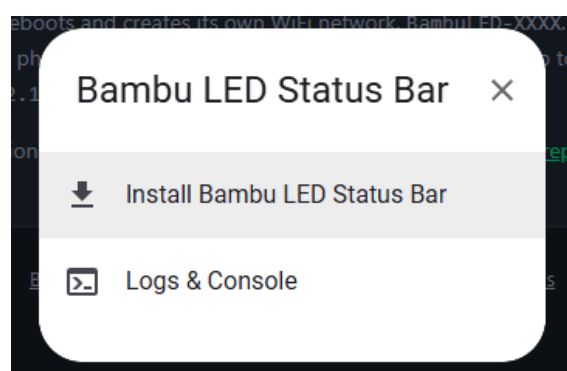
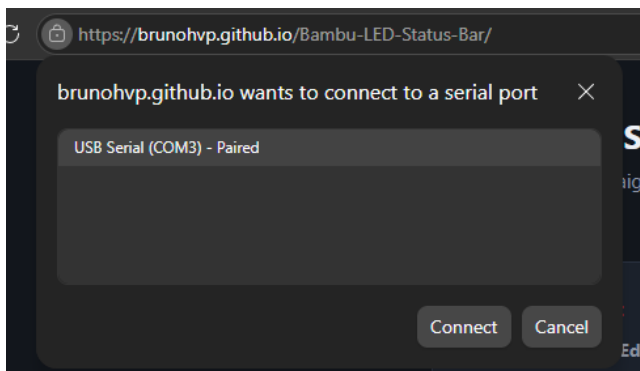
Open the following website using Chrome or Edge:

[Bambu LED Status Bar – Firmware Installer](https://brunohvp.github.io/Bambu-LED-Status-Bar/)

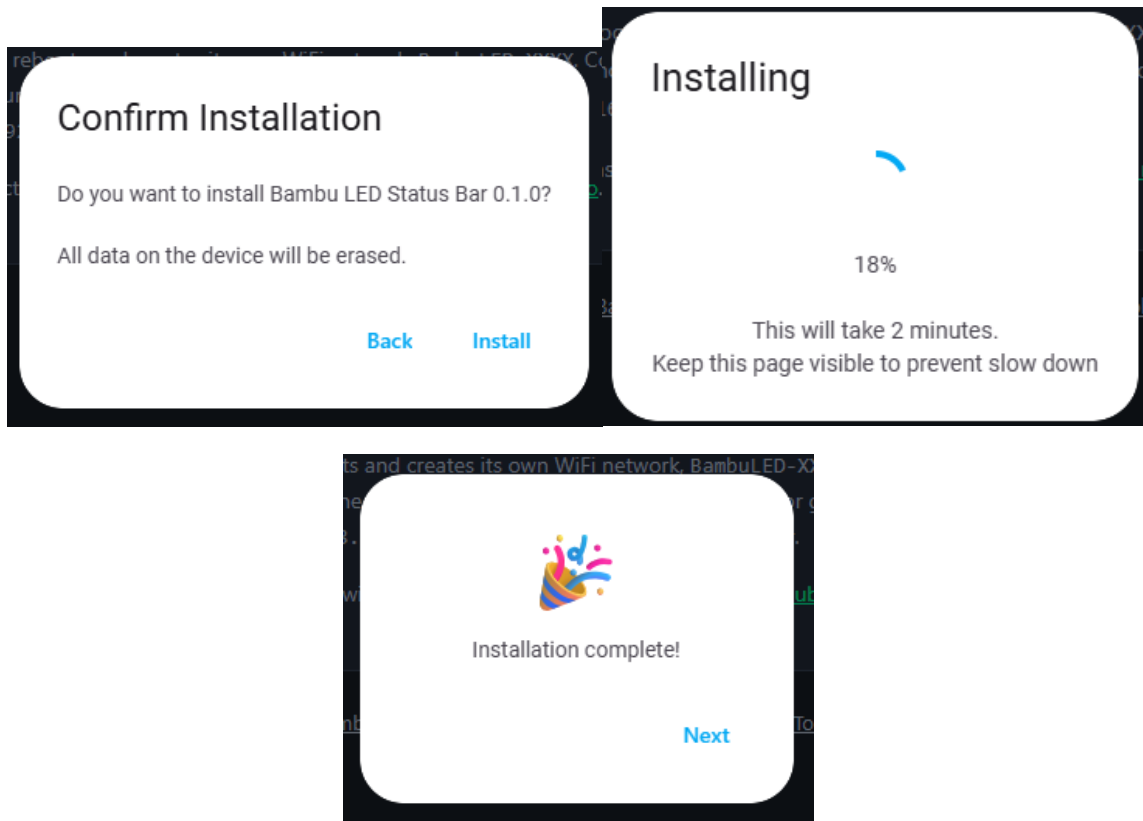
Connect your ESP32 to your computer using a USB data cable, then follow the instructions on the website to install the firmware.



Click **Connect/Install**. Your browser asks which serial port to use — pick the ESP32's (shows up as "USB Serial" or similar).



Confirm the install, then just wait — it flashes and verifies on its own, no file to choose. Keep the tab open/visible while it works (it says why).



Once installation is complete, disconnect and reconnect the ESP32 if necessary.

Important: Flashing the ESP32 may erase any existing firmware and configuration stored on the board.

2. Connecting to Wi-Fi

When the ESP32 starts for the first time, it creates its own Wi-Fi network named BambuLED-XXXX.

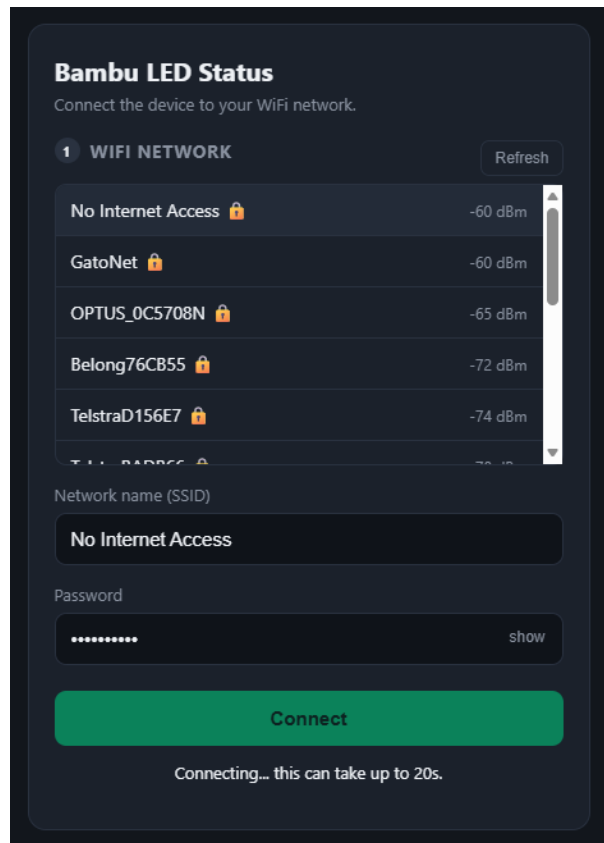
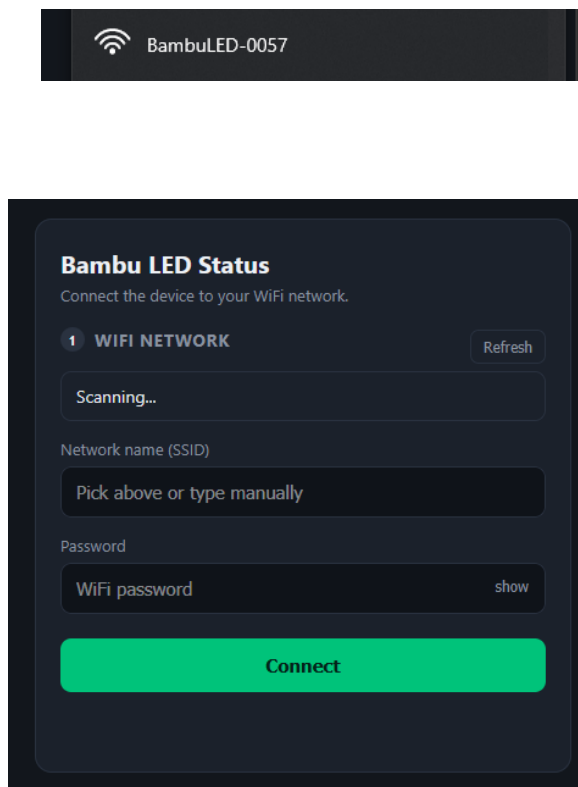
Connect to this network using your phone or computer. The setup page should open automatically. If it doesn't, open your browser and navigate to:

<http://192.168.4.1>

On the setup page, select your home Wi-Fi network and enter its password.

Click Connect and wait for the ESP32 to establish a connection.

The setup portal remains available while the ESP32 connects to your network, allowing you to continue configuring your printer.



3. Configuring Your Bambu Lab Printer

Before connecting the ESP32 to your printer, make sure LAN Mode is enabled. The standalone firmware uses MQTT to communicate directly with the printer over your local network.

Enable LAN Mode

1. On your printer's touchscreen, go to Settings → WLAN → LAN Only Mode.
2. Enable LAN Only Mode.
3. Open the information/settings icon next to LAN Mode to find your LAN Access Code and Serial Number.
4. Make sure the printer is connected to your local network. You'll also need its IP address if automatic discovery doesn't work.

Connect Your ESP32 to the Printer

After completing the Wi-Fi setup, the ESP32 will automatically start searching for Bambu Lab printers on your network.

Step 1 – Find your printer

If your printer appears in the list, simply select it. The printer's IP address and serial number should be filled in automatically.

The image shows two side-by-side screenshots of a web-based configuration interface for a printer. Both screenshots are on a dark background with light text and green accents.

Left Screenshot:

- At the top, it says "CONNECTED" with a green checkmark icon. Below it, "Joined 'No Internet Access' — IP 10.0.20.197".
- The main heading is "2 FIND YOUR PRINTER".
- Below that, it says "Found 1 printer(s):" followed by a "Search again" button.
- A card displays "BrPrint (N7)" and "10.0.20.40".
- A link "Enter printer info manually" is shown in green.
- Fields for "Printer IP" (with example "e.g. 192.168.1.50"), "Serial number" (with example "e.g. 01P00A0000000000"), and "Access Code" (with "LAN Mode access code" and a "show" button) are present.
- A green "Save & finish" button is at the bottom.
- At the very bottom, a note reads: "On the printer Settings > WLAN > enable LAN Only Mode > tap the gear/info icon to see the Access Code and Serial Number."

Right Screenshot:

- Similar to the left, but the "Access Code" field is highlighted with a green border, and the "show" button is more prominent.
- The "Save & finish" button is also highlighted with a green border.
- The note at the bottom is the same as the left screenshot.

Step 2 – Enter your LAN Access Code

Enter the LAN Access Code displayed on your printer. The printer doesn't broadcast this information, so you'll need to enter it manually.

If your printer doesn't appear, click Search again. If automatic discovery still doesn't work, select Enter printer info manually and provide the printer's IP address, serial number and LAN Access Code.

Save and Connect

Once all the details are entered, save your configuration.

The ESP32 will store the settings and reboot, connecting to your printer over MQTT. The temporary BambuLED-XXXX Wi-Fi network will disappear once setup is complete.

You can then access the LED Status Bar web interface at:

`http://bambuled.local`

Alternatively, use the ESP32's IP address if the hostname doesn't work on your network.

The web interface will display the printer's current status, MQTT connection status and a live preview of the LED bar.

4. Web Interface & LED Customisation

Once your ESP32 is connected to the printer, you can access its web interface from any device on your local network.

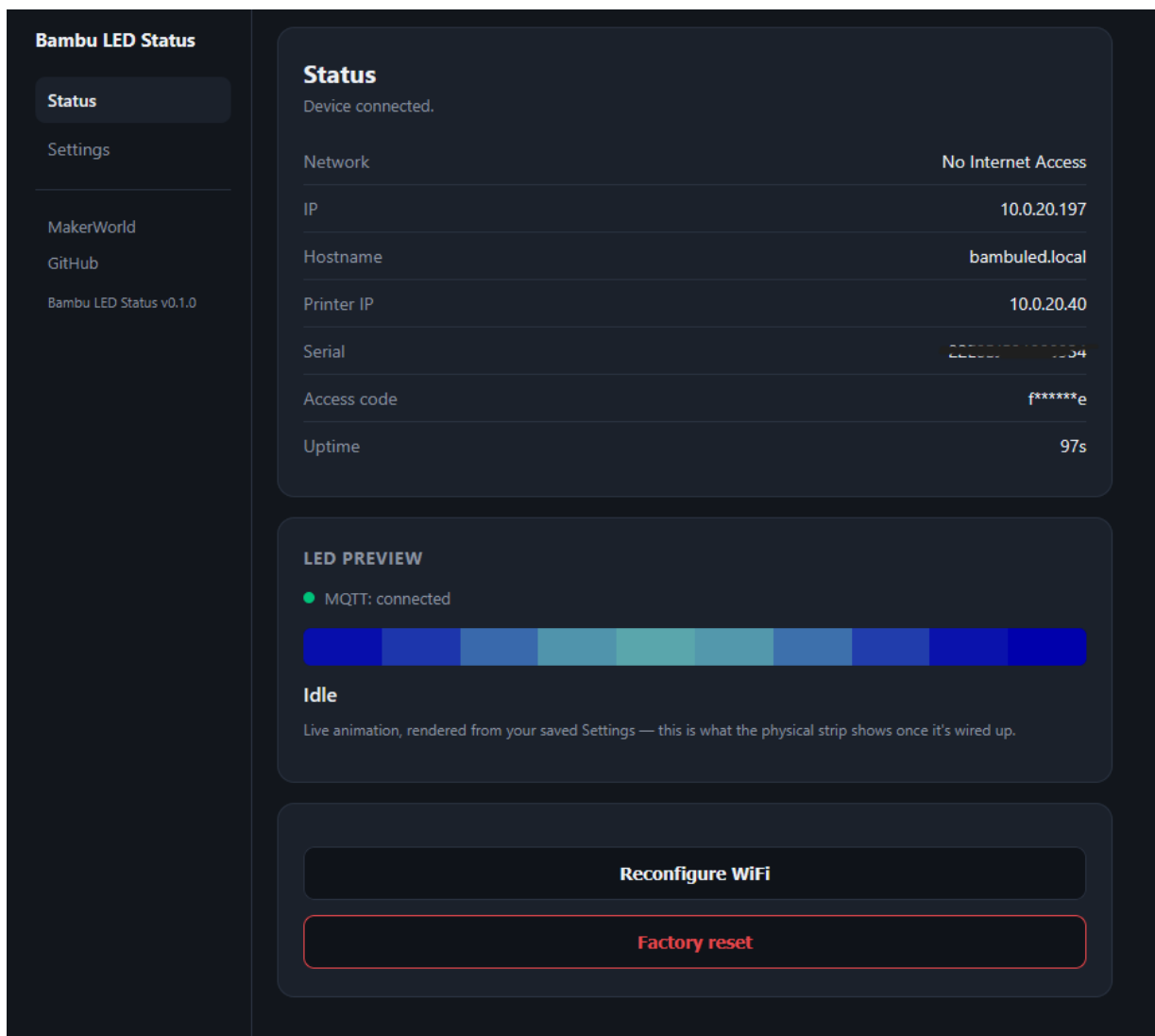
Open your browser and go to:

<http://bambuled.local>

If that address doesn't work, use the ESP32's IP address instead.

Live Status

The main page displays the printer's current status, Wi-Fi and MQTT connection information, and a live preview of the LED strip.



You can use this page to check whether your ESP32 is communicating with the printer without connecting it to your computer.

LED Settings

To customise your LED bar, click on settings:

Bambu LED Status

StatusSettingsMakerWorldGitHubBambu LED Status v0.1.0

LED ANIMATIONS

Pick colors per state — the animation itself is fixed. Brightness above applies to the whole strip. Changes preview live below, then hit Save.

Brightness (whole strip, all states)

Idle

Color AColor B

#85F5FF

#0000FF

Live preview

Calibrating

Color

#0000FF

Live preview

Printing

Bar ColorLeading Pixel

#00AEEF

#FFFFFF

Live preview

Paused

Color

#FF3C00

Live preview

Finished

Color

#00FF44

Live preview

Error

Color

#FF0000

Live preview

Restore Defaults

Save

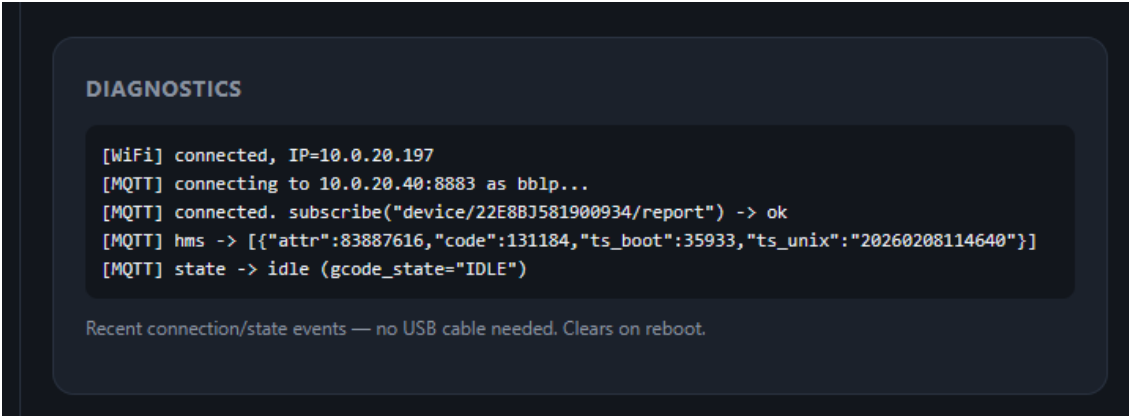
Each printer state has its own animation and configurable colours. You can change the colours and overall brightness directly from the web interface without editing the firmware or reflashing your ESP32.

The animation types are already configured, so you only need to choose the colours you prefer.

Note: The colours displayed in the web interface may look slightly different on your actual LED strip. Adjust them until you're happy with the result.

Diagnostics

The Settings page also includes a Diagnostics panel displaying recent firmware activity.



5. LED Animations & Printer States

The firmware automatically changes the LED animation according to the printer's current state.

Printer state	Animation	Behaviour
Idle	Plasmoid	A slow colour wave along the strip
Calibrating	Loading	A moving LED with a fading trail
Printing	Percent	A progress bar that grows as the print progresses
Paused	Fade	Slow breathing animation
Finished	Fade	Green breathing animation
Error	Fade	Fast breathing animation

Heating, homing, bed levelling and filament changes are all included under Calibrating.

The default configuration uses 10 controllable LED sections, with each section representing approximately 10% of print progress.

Automatic Brightness

The firmware automatically adjusts the brightness based on the printer's chamber light.

When the chamber light is switched off, the LED bar dims to approximately 10% of its configured brightness.

If the printer remains idle for 30 minutes, the LEDs dim further.

Finished Prints

After a print finishes, the LED bar displays the Finished animation for approximately 60 seconds before returning to Idle.

This prevents the LED bar from staying green indefinitely if the printer doesn't immediately report that it has returned to Idle.

Lost Connection

If the ESP32 temporarily loses its MQTT connection, the LED strip keeps displaying its last known state while the firmware attempts to reconnect.

It won't switch to a flashing error animation just because of a brief network interruption.

6. Testing Your Setup

After configuring the printer and adjusting the LED colours, I recommend running a short test print.

Check that the LED bar changes from Calibrating to Printing and that the progress bar advances during the print.

You can also pause and resume the print to test the Paused animation.

When the print finishes, the LEDs should display the Finished animation and automatically return to Idle after approximately 60 seconds.

If you want to test the Error animation, use an error condition that you can safely reproduce and clear. There's no need to deliberately interrupt moving printer components.

7. Resetting the ESP32

If you need to change your Wi-Fi network or start the setup again, the main status page provides two reset options.

Reconfigure WiFi clears the saved Wi-Fi credentials while keeping your printer configuration. The ESP32 returns to setup mode so you can connect it to another wireless network.

Factory reset clears the saved configuration and starts the entire setup process again, including printer connection details and custom LED settings.

Known Issue / Note:

This firmware is still a beta project, and I'm continuing to make improvements based on testing and community feedback.

Occasional ESP32 Restarts - The current firmware may occasionally restart following a TLS/MQTT connection error.

It automatically reconnects to Wi-Fi and the printer, and your saved configuration is retained. The LED strip keeps its previous state during brief reconnection interruptions.

Printer Error State - After a cancelled or failed print, some Bambu printers may continue reporting a failed state even after the error has been dismissed.

If the LED bar remains in Error after the issue has been resolved, restarting the printer may be necessary.

For technical details and the latest updates on these issues, check the [GitHub README](#).

Project Links

For the latest version of the project, updates, files and community feedback, visit the MakerWorld project page:

Bambu Lab P2S LED Progress / Status Bar

[MakerWorld Project Page](#)

If you build this project, adapt it to another Bambu Lab printer, or improve the automation, feel free to share your results in the comments.

I'll keep updating the project as I improve the firmware, fix reported issues and test compatibility with more Bambu Lab printers.

CAD / Mount Source

You can also modify the LED holder using the original Onshape design:

[Onshape Document](#)

Thanks for checking out the project, and have fun building your own LED status bar. 😊